

IRELE

Advanced Measurement and Data Driven Modelling

Lab report

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Lab 1: RLS with forgetting factor

1.1 Main steps

1. Measure the input and output signals from the DUT. The input is arbitrary and the system is time varying.
- 2.

1.2 Time-varying RLS estimation

The system's equation can be rewritten in a vector form as follows:

$$\begin{aligned} y(N) &= b_0(N)u(N) + b_1(N)u(N-1) - a_1(N)y(N-1) - a_2(N)y(N-2) \\ &= K(N)\theta(N) \end{aligned}$$

with

$$K(N) = \begin{bmatrix} u(N) & u(N-1) & -y(N-1) & -y(N-2) \end{bmatrix}, \quad \theta(N) = \begin{bmatrix} b_0(N) \\ b_1(N) \\ a_1(N) \\ a_2(N) \end{bmatrix}$$

The first estimate for θ and P are determined using a least squares approach on the first 200 samples. The RLS is then applied to the rest of the data.

The Matlab code is given below:

```
% initialization with an LLS estimate over a few of the first samples
% y_lls = H * theta
H = zeros(sampleForLLS - max(na, nb), na + nb + 1);
for uCol = 0:nb
```

```

        H(:, uCol + 1) = u((max(na, nb) + 1 - uCol:sampleForLLS - uCol));
    end
    for yCol = 1:na
        H(:, nb + 1 + yCol) = -y((max(na, nb) + 1 - yCol:sampleForLLS - yCol));
    end
    y_lls = y((max(na, nb) + 1:sampleForLLS));

    theta = H\y_lls;
    P = inv(H'*H);

    theta_sav = zeros(na + nb + 1, N - max(na, nb) + 2);
    P_diag_sav = zeros(na + nb + 1, N - max(na, nb) + 2);

    for itr = sampleForLLS+1:N

        % saving previous values
        theta_sav(:, itr - max(na, nb) + 1) = theta;
        P_diag_sav(:, itr - max(na, nb) + 1) = diag(P);

        K_itr = [u(itr - (0:nb)), -y(itr - (1:na))];

        theta = theta - (P * (K_itr') * (K_itr * theta - y(itr))) / (g + K_itr * P * (K_itr'));
        P = 1/g * (P - (P * (K_itr') * K_itr * P) / (g + K_itr * P * (K_itr')));

    end

```